PROJECT CASE STUDY

# Singhar By Sidra

*An editorial-first fashion storefront connecting occasion-led discovery, Shopify commerce, and customer accounts through one considered shopping experience.*

---

SYSTEM

**Headless Fashion Storefront**

SCOPE

**Discovery · Commerce Customer Accounts**

APPROACH

**Server-Rendered Shopify + Editorial CMS**

## 01 The Problem

A fashion storefront needs to communicate the character of a brand without compromising the clarity and reliability of shopping.

Luxury eastern wear calls for editorial storytelling, collection-led discovery, and careful product presentation. Behind that experience, products, variants, cart state, checkout, and customer records must stay connected to a dependable commerce system.

> Singhar By Sidra brings brand-led merchandising and Shopify-owned commerce into one storefront — with a separate editorial layer and authenticated customer journeys.

## 02 What Was Built

A connected shopping experience spanning editorial discovery, product evaluation, cart, checkout handoff, and post-purchase account tasks.

### EDITORIAL HOMEPAGE

Combine BILDIT-managed hero and promotional slots with authored fallbacks, featured collections, new arrivals, and best sellers.

### CATALOG & DISCOVERY

Browse collections with filtering, sorting, and pagination; find products through regular and predictive search and select product variants.

### CART & CHECKOUT

Share cart functionality across a drawer and dedicated page, with optimistic feedback, discount handling, and Shopify-hosted checkout.

### CUSTOMER EXPERIENCE

Connect authenticated profiles, addresses, order history, and saved products, alongside an integrated product review experience.

# 03 How It Works

Hydrogen and React Router serve the storefront on Shopify Oxygen. Server loaders and actions connect commerce, customer identity, and editorial content; React adds focused browser interactions.

**Browser → Oxygen worker → Hydrogen + React Router**

Storefront API → Catalog · Search · Cart · Shopify checkout

Customer Account API → Accounts · Wishlist | BILDIT → Homepage slots

### Server-owned data & streaming

Each request receives a Hydrogen context for Shopify clients, cart, session, cache, and the English / Pakistan market context. Critical content is awaited while supporting data streams through Suspense and Await.

### Editorial content, commerce boundaries

Shopify remains the source of truth for products and transactions. BILDIT supplies homepage banners and replaceable hero/promo slots; authored fallbacks preserve the page when CMS content is unavailable.

### Cart mutations & checkout handoff

React Router forms and fetchers submit operations to the shared cart action. Hydrogen manages cart identity and optimistic state; checkout continues at Shopify's supplied checkout URL.

### Customer identity & product confidence

Shopify Customer Account OAuth protects account tasks and saved products. The wishlist uses a customer metafield, while Judge.me provides product reviews. Personalized account data is not shared-cache content.

## 04 Key Features

**Brand-led merchandising** — an editorial homepage combines campaign storytelling, featured collections, new arrivals, and best sellers.

**Flexible homepage content** — BILDIT-managed hero and promo slots sit alongside storefront-owned fallbacks without replacing Shopify commerce data.

**Collection discovery** — Shopify-backed filters, sorting, and cursor pagination help shoppers explore the catalog.

**Regular and predictive search** — dedicated search paths support both full results and in-context product discovery.

**Variant-aware product pages** — Hydrogen product-option helpers connect selected options, product URLs, and the add-to-cart journey.

**Connected cart experience** — a cart drawer and cart page share Hydrogen cart behavior, optimistic updates, and checkout handoff.

**Authenticated customer accounts** — OAuth-backed profile, address, and order pages support post-purchase self-service.

**Saved products and reviews** — customer wishlists retain product references, while Judge.me adds product review display and submission.

**Edge-compatible rendering** — Oxygen architecture, streamed server rendering, and nonce-based CSP preserve clear runtime and security boundaries.

# 05 Technical Approach

The implementation separates storefront presentation, server-owned commerce, customer identity, and editorial management. The following stack supports those boundaries.

| LAYER | STACK | PURPOSE |
|---|---|---|
| Application | Hydrogen 2026.4.5; React 18; React Router 7; strict TypeScript | Typed routes, loaders, actions, and interactive components |
| Commerce | Shopify Storefront API; Hydrogen cart and variant primitives | Catalog, search, product options, cart, and checkout handoff |
| Customer identity | Shopify Customer Account API; OAuth; customer metafields | Accounts, orders, addresses, and customer wishlists |
| Editorial & reviews | BILDIT Hydrogen adapter; Judge.me integration | Managed homepage slots and product review workflows |
| Presentation | Tailwind CSS v4; local Inter and Playfair Display fonts | Responsive storefront styling and brand typography |
| Runtime & tooling | Shopify Oxygen; Vite 8; GraphQL codegen; TypeScript; ESLint | Streaming worker runtime, generated API types, and static checks |

## Design decisions that matter

**Keep ownership explicit.** Shopify owns commerce and customer records; BILDIT manages editorial presentation. The storefront does not recreate checkout or maintain a parallel browser cart.

**Load by priority.** Critical route data and deferred sections use distinct loading paths, so supporting content does not unnecessarily block the initial page.

**Separate public and private data.** Catalog content and personalized account state have different caching needs. Server-side sessions and API clients keep sensitive operations outside browser code.

## 06 Value Delivered

From brand storytelling to a connected, Shopify-backed shopping journey:

### Editorial

BRAND & MERCHANDISING

### Connected

DISCOVERY & COMMERCE

### Customer-led

ACCOUNTS & SAVED PRODUCTS

- **A clearer path from inspiration to purchase.** — Homepage storytelling, collections, search, and product pages connect editorial interest to a Shopify-owned cart and checkout.
- **Editorial flexibility without commerce duplication.** — Managed homepage slots let content evolve independently while catalog, pricing, variants, and transactions remain owned by Shopify.
- **Continuity beyond the product page.** — Saved products, customer profiles, addresses, and order history keep discovery and post-purchase tasks within the same storefront.
- **Clearer implementation boundaries.** — Server loaders and actions, typed API operations, and focused integrations give commerce, editorial content, and identity defined responsibilities.

*Singhar By Sidra — an editorial-first fashion storefront bringing Shopify commerce, managed homepage content, and connected customer experiences into one application.*